

LECTURE#06

MULTIVARIABLE CALCULUS

LAPLACE'S EQUATION

For a function $w = f(x, y, z)$ then equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0,$$

Is called Laplace's equation in 3-variables.

Example 01:

$$f(x, y) = e^x \sin y + e^y \cos x$$

Solution:

$$\frac{\partial f}{\partial x} = e^x \sin y - e^y \sin x$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (e^x \sin y - e^y \sin x) = e^x \sin y - e^y \cos x$$

CONT...

$$\frac{\partial f}{\partial y} = e^x \cos y + e^y \cos x$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (e^x \cos y + e^y \cos x) = -e^x \sin y + e^y \cos x$$

Adding both partial second order derivatives, we have

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^x \sin y - e^y \sin x - e^x \sin y + e^y \cos x = 0$$

EXAMPLE 02

$$f(x, y) = \frac{x + y}{x - y}$$

Solution:

$$f_x(x, y) = \frac{(x-y)\frac{\partial}{\partial x}(x+y) - (x+y)\frac{\partial}{\partial x}(x-y)}{(x-y)^2}$$

$$= \frac{(x-y)(1) - (x+y)(1)}{(x-y)^2}$$

$$f_x(x, y) = \frac{-2y}{(x-y)^2}$$

CONT...

$$\begin{aligned} f_y(x, y) &= \frac{(x - y) \frac{\partial}{\partial y} (x + y) - (x + y) \frac{\partial}{\partial y} (x - y)}{(x - y)^2} \\ &= \frac{(x - y)(1) - (x + y)(-1)}{(x - y)^2} \\ f_x(x, y) &= \frac{2x}{(x - y)^2} \end{aligned}$$

EXAMPLE 03

$$f(x, y) = x^3 e^{-y} + y^3 \sec \sqrt{x}$$

Solution:

$$f_x(x, y) = 3x^2 e^{-y} + y^3 \sec \sqrt{x} \cdot \tan \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$$

$$= 3x^2 e^{-y} + \frac{y^3 \sec \sqrt{x} \cdot \tan \sqrt{x}}{2\sqrt{x}}$$

$$f_y(x, y) = -x^3 e^{-y} + 3y^2 \sec \sqrt{x}$$

EXAMPLE 04

$$f(x, y) = x^2 y e^{xy}$$

Solution:

$$\begin{aligned} f_x(x, y) &= 2xye^{xy} + x^2 y^2 e^{xy} \\ &= xye^{xy}(2 + xy) \end{aligned}$$

$$f_x(1, 1) = (1)(1)e^{(1)(1)}(2 + (1)(1))$$

$$f_x(1, 1) = 3e$$

$$\begin{aligned} f_y(x, y) &= x^2 e^{xy} + x^3 y e^{xy} \\ &= x^2 e^{xy}(1 + xy) \end{aligned}$$

$$f_y(1, 1) = (1)^2 e^{(1)(1)}(1 + (1)(1))$$

$$f_y(1, 1) = 2e$$

EXAMPLE 05

$$f(x, y) = x^2 \cos(xy)$$

Solution: $f_x(x, y) = 2x \cos(xy) - x^2 y \sin(xy)$

$$f_x\left(\frac{1}{2}, \pi\right) = 2\left(\frac{1}{2}\right) \cos\left(\left(\frac{1}{2}\right)(\pi)\right) - \left(\frac{1}{2}\right)^2 (\pi) \sin\left(\left(\frac{1}{2}\right)(\pi)\right) = -\frac{\pi}{4}$$

$$f_y(x, y) = -x^3 \sin(xy)$$

$$f_y\left(\frac{1}{2}, \pi\right) = -\left(\frac{1}{2}\right)^3 \sin\left(\left(\frac{1}{2}\right)(\pi)\right) = -\frac{1}{8}$$

EXERCISE

1. Let $f(x, y, z) = x^3y^5z^7 + xy^2 + y^3z$. Find $f_{xy}, f_{yx}, f_{xz}, f_{zz}, f_{zyy}, f_{xxy}, f_{zyx}, f_{xxyz}$.
2. Let $w(x, y, z) = (4x - 3y + 2z)^5$. Find $\frac{\partial^2 w}{\partial x \partial z}, \frac{\partial^3 w}{\partial x \partial y \partial z}, \frac{\partial^4 w}{\partial z^2 \partial y \partial x}$.
3. Let $f(x, y) = (e)^{x-y^2}$. Find $f_{xx}, f_{yy}, f_{xy}, f_{yx}$.
4. Given $f(x, y) = y^3e^{-5x}$. Find $f_{xyy}(0,1), f_{xxx}(0,1), f_{yyxx}(0,1)$.

5. Given $z = \sqrt{x^2 + 4y^2}$ Find $\frac{\partial z}{\partial x}\bigg|_{(1,2)}, \frac{\partial z}{\partial y}\bigg|_{(1,2)}$
6. Let $w = \sqrt{x^2 + 4y^2 - z^2}$ Find $\frac{\partial w}{\partial x}\bigg|_{(2,1,-1)}, \frac{\partial w}{\partial y}\bigg|_{(2,1,-1)}, \frac{\partial w}{\partial z}\bigg|_{(2,1,-1)}$
7. Let $w = x \sin(xy)$ Find $\frac{\partial w}{\partial x}\bigg|_{(1, \frac{1}{2}, \pi)}, \frac{\partial w}{\partial y}\bigg|_{(1, \frac{1}{2}, \pi)}, \frac{\partial w}{\partial z}\bigg|_{(1, \frac{1}{2}, \pi)}$

EXERCISE

A function f of x and y is harmonic if $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$. Prove that the functions defined in Exercises 1-5 are harmonic.

1) $f(x, y) = \ln(x^2 + y^2)$

2) $f(x, y) = \tan^{-1} \left(\frac{2xy}{x^2 - y^2} \right)$

3) $f(x, y) = \arctan \left(\frac{y}{x} \right)$

4) $f(x, y) = \cos(x) \sinh(y) + \sin(x) \cosh(y)$

5) $f(x, y) = e^{-x} \cos y + e^{-y} \cos x$

EXERCISE

6. If $w = \cos(x - y) + \ln(x + y)$, show that $\frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2} = 0$.
7. If $w = (y - 2x)^2 - \sqrt{y - 2x}$, show that $w_{xx} - 4w_{yy} = 0$.
8. If $w = e^{-c^2 t} \sin(cx)$, show that $w_{xx} = w_t$ for every real number c .

In Exercises 9 and 10 show that v satisfies the wave equation

$$\frac{\partial^2 v}{\partial t^2} = a^2 \frac{\partial^2 v}{\partial x^2}.$$

9. $v = (\sin(akt))(\sin(kx))$
10. $v = (x - at)^4 + \cos(x + at)$

Two functions u and v of x and y are said to satisfy the *Cauchy–Riemann equations* if $u_x = v_y$ and $u_y = -v_x$. Show that the pairs of functions in Exercises 43–46 satisfy these equations.

43 $u(x, y) = x^2 - y^2, v(x, y) = 2xy$

44 $u(x, y) = y/(x^2 + y^2), v(x, y) = x/(x^2 + y^2)$

45 $u(x, y) = e^x \cos y, v(x, y) = e^x \sin y$

46 $u(x, y) = \cos x \cosh y + \sin x \sinh y$
 $v(x, y) = \cos x \cosh y - \sin x \sinh y$

- 51** A flat metal plate is situated on an xy -plane such that the temperature T at the point (x, y) is given by $T = 10(x^2 + y^2)^2$, where T is in degrees and x, y are in centimeters. Find the rate of change of T with respect to distance at $P(1, 2)$ in the direction of (a) the x -axis; (b) the y -axis.
- 52** The surface of a certain lake is represented by a region D in the xy -plane such that the depth under the point corresponding to (x, y) is given by $f(x, y) = 300 - 2x^2 - 3y^2$, where x, y and $f(x, y)$ are in feet. If a boy in the water is at the point $(4, 9)$, find the rate at which the depth changes if he swims in the direction of (a) the x -axis; (b) the y -axis.
- 53** Suppose the electrical potential V at the point (x, y, z) is given by $V = 100/(x^2 + y^2 + z^2)$, where V is in volts and x, y, z are in inches. Find the rate of change of V with

respect to distance at $P(2, -1, 1)$ in the direction of (a) the x -axis; (b) the y -axis; (c) the z -axis.

- 54** An object is situated in a rectangular coordinate system such that the temperature T at the point $P(x, y, z)$ is given by $T = 4x^2 - y^2 + 16z^2$, where T is in degrees and x, y, z are in centimeters. Find the rate of change of T at the point $P(4, -2, 1)$ in the direction of (a) the x -axis; (b) the y -axis; (c) the z -axis.
- 55** Let C be the trace of the paraboloid $z = 9 - x^2 - y^2$ on the plane $x = 1$. Find parametric equations of the tangent line l to C at the point $P(1, 2, 4)$. Sketch the paraboloid, C , and l . (*Hint*: See Figure 16.17.)
- 56** Let C be the trace of the graph of $z = \sqrt{36 - 9x^2 - 4y^2}$ on the plane $y = 2$. Find parametric equations of the tangent line l to C at the point $(1, 2, \sqrt{11})$. Sketch the surface, C , and l .